GridMind AI

AI-Powered Renewable Electricity Intelligence & Optimisation Platform for MSMEs

AI Builder Cup — Final Proposal & Architecture Specification

1. Executive Summary

Electricity procurement is typically the 2nd highest operational expense in manufacturing MSMEs, representing 15% to 35% of total operating budgets. Complex multi-tier Time-of-Day (ToD) tariffs, uncoordinated shift startups, and underutilised on-site solar, wind, and biomass result in heavy recurring losses. **GridMind AI** combines deterministic industrial physics, linear programming (SciPy HiGHS solver), and grounded Google Gemini AI to help MSME plant managers understand electricity waste, orchestrate clean energy sources, and reduce estimated procurement costs by 20% to 45%.

2. System Architecture & Technology Stack

Layer	Technology	Role & Boundary
Frontend UI	React 18, Vite, TypeScript, Tailwind CSS, Recharts	Factory switcher, real-time telemetry, LP dispatch charts, alerts
Backend Core	Python 3.11+, FastAPI, Pydantic v2	Deterministic physics simulator, REST APIs, ToD tariff modeling
Optimization	SciPy HiGHS Solver (LP simplex & interior-point)	Constrained cost-minimizing multi-source hourly energy allocation
Generative AI	Google Gemini 2.0 (via Google GenAI SDK)	Evidence-grounded anomaly investigation & executive reporting
Cloud Deploy	Google Cloud Run & Firebase Hosting	Containerized autoscaling microservices with Secret Manager

CRITICAL ARCHITECTURAL BOUNDARY: *The Large Language Model (Gemini) never calculates rupee balances, consumption totals, or dispatch limits. All numbers are computed authoritatively by the deterministic backend and supplied to Gemini as verified evidence.*

3. Three Configured Demo MSME Factory Profiles

GridMind AI supports exactly three distinct Indian MSME factory archetypes with isolated datasets, distinct connected loads, and tailored generation profiles:

Factory ID & Name	Location & Industry	Connected Load / Grid Limit	Renewable Assets	Key Demonstration Scenario
FAC-IN-PUNE-001 Apex Precision	Chakan, Pune, MH CNC & Auto Components	260 kW Load 250 kW Contract Demand	100 kWp Solar 60 kW Wind 50 kW Biomass	Detect off-shift standby compressor waste (0
FAC-IN-NASHIK-002 FreshHarvest Foods	Satpur, Nashik, MH Food & Cold Storage	210 kW Load 200 kW Contract Demand	75 kWp Solar 20 kW Wind 35 kW Agro-Biomass	Continuous refrigeration demand during even
FAC-IN-TIRUPPUR-003 LoomCraft Textiles	Tiruppur, TN Spinning & Weaving	280 kW Load 270 kW Contract Demand	120 kWp Solar 50 kW Wind 45 kW Pellet Biomass	High solar self-consumption and wind/biomas

4. Linear Programming Dispatch Formulation (HiGHS)

Objective: Minimize total daily electricity procurement cost across 24 hourly intervals:

- $\min \Sigma_t [c_{solar} \cdot x_{solar,t} + c_{wind} \cdot x_{wind,t} + c_{biomass} \cdot x_{biomass,t} + Tariff_t \cdot x_{grid,t}]$

Physical Constraints Enforced on Every Interval:

1. **Energy Balance:** $x_{solar,t} + x_{wind,t} + x_{biomass,t} + x_{grid,t} = Demand_t$
2. **Solar Resource Cap:** $0 \le x_{solar,t} \le \min(SolarAvailable_t, SolarCapacity)$
3. **Wind Resource Cap:** $0 \le x_{wind,t} \le \min(WindAvailable_t, WindCapacity)$
4. **Biomass Power & Fuel Budget:** $0 \le x_{biomass,t} \le Capacity$ and $\Sigma\, x_{biomass,t} \le DailyBudgetKg / SFC$
5. **Sanctioned Grid Headroom:** $0 \le x_{grid,t} \le ContractDemandLimit$

5. Google Gemini Grounded Intelligence

Gemini provides 5 production capabilities without hallucinating calculations: (1) **Single Anomaly Root-Cause Investigation** with structured hypotheses; (2) **Prioritized Action Recommendations** with safety checks; (3) **Mathematical Optimization Explanation** clarifying why sources were dispatched; (4) **Full Factory Executive Briefing** synthesizing plant-wide energy risks; and (5) **Context-Aware Operator Q&A;**.

6. Quality Verification & Test Coverage

• **Backend Tests**: 97 automated pytest tests passing in ~10 seconds.

• **Frontend Build**: Zero TypeScript errors; production bundle built cleanly in ~19s.

• **Data Isolation**: Full per-factory cache isolation verified across all endpoints.

• **Disclaimer**: All factory readings, equipment telemetry, and savings figures are synthetically generated for prototype evaluation.